SAGAR PUBLIC SCHOOL

Integrated Programme | Topic: Current Electricity – Level 3

PRACTICE SHEET – LEVEL 3

Current Electricity – Power, Drift, Bridge, Cells & Kirchhoff's Laws

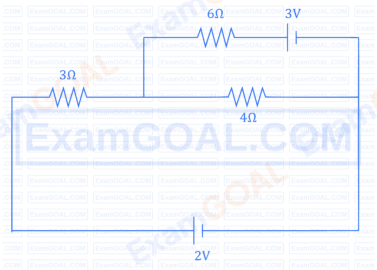
Time Allowed: 60 Mins **Total Questions: 30**

Name: _____ Class/Sec: _____ Date: _____

Instructions: Attempt all questions. Each question with options carries one correct option. Questions without options are numerical-value type: write the answer in the blank. The tag after each question shows its source examination. A self-assessment page follows; the answer key is provided as a separate sheet.

Q.1 A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J. *[JEE Main 2022, 25 Jun, Mor.]*

Q.2 Refer to the circuit diagram given below. The heat generated across the $6\,\Omega$ resistance in 100 second is $\frac{\alpha}{100}$ J. The value of α is _____. (Nearest integer) *[JEE Main 2026, 6 Apr, Mor.]*



Q.3 An electrical bulb rated 220 V, 100 W, is connected in series with another bulb 220 V, 60 W. If the voltage across combination is 220 V, the power consumed by the 100 W bulb will be about _____ W. *[JEE Main 2022, 28 Jul, Eve.]*

Q.4 A uniform heating wire of resistance $36\,\Omega$ is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240 V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be $1 : x$, where x is _____. *[JEE Main 2021, 1 Sep, Eve.]*

Q.5 A battery with EMF E and internal resistance r is connected across a resistance R. The power consumption in R will be maximum when: *[JEE Main 2026, 21 Jan, Eve.]*

(A) $R = \sqrt{2}\,r$ (B) $R = 2r$

(C) $R = \frac{r}{2}$ (D) $R = r$

Q.6 The battery of a mobile phone is rated as 4.2 V, 5800 mAh. How much energy is stored in it when fully charged? *[JEE Main 2025, 2 Apr, Mor.]*

(A) 87.7 kJ (B) 43.8 kJ

(C) 48.7 kJ (D) 24.4 kJ

Q.7 The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t = 1$ s to $t = 2$ s is _____ C. *[JEE Main 2024, 1 Feb, Mor.]*

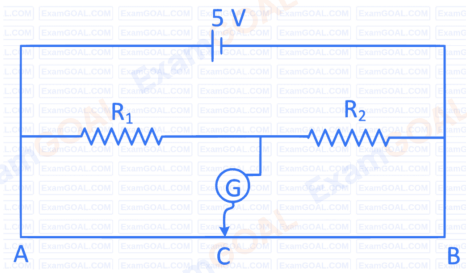
Q.8 A cylindrical conductor of length 2 m and area of cross-section 0.2 mm^2 carries an electric current of 1.6 A when its ends are connected to a 2 V battery. Mobility of electrons in the conductor is $\alpha \times 10^{-3}$ $m^2/V \cdot s$. The value of α is: (electron concentration $= 5 \times 10^{28}/m^3$ and electron charge $= 1.6 \times 10^{-19}$ C) *[JEE Main 2026, 22 Jan, Eve.]*

Q.9 The current density in a cylindrical wire of radius $r = 4.0$ mm is 1.0×10^6 A/m^2. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi$ A; where x is _____. *[JEE Main 2022, 27 Jun, Eve.]*

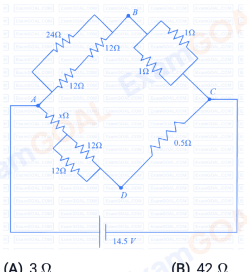
Q.10 In a meter bridge experiment to determine the value of unknown resistance, first the resistances $2\,\Omega$ and $3\,\Omega$ are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance $x\,\Omega$ is connected in parallel to $3\,\Omega$ resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is _____ Ω. *[JEE Main 2026, 24 Jan, Eve.]*

Q.11 When two resistance R_1 and R_2 connected in series and introduced into the left gap of a meter bridge and a resistance of $10\,\Omega$ is introduced into the right gap, a null point is found at 60 cm from left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of $3\,\Omega$ is introduced into the right gap to get null point at 40 cm from left end. The product of R_1R_2 is _____ Ω^2. *[JEE Main 2023, 29 Jan, Eve.]*

Q.12 In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is 40 cm. The balancing length, if the radius of the wire AB is doubled, will be _____ cm. *[JEE Main 2022, 27 Jul, Eve.]*



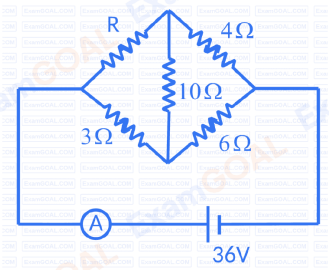
Q.13 The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is: *[JEE Main 2024, 6 Apr, Mor.]*



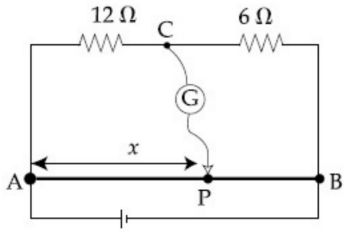
(A) $3\,\Omega$ (B) $42\,\Omega$ (C) $6\,\Omega$ (D) $9\,\Omega$

Q.14 Four resistances of $15\,\Omega$, $12\,\Omega$, $4\,\Omega$ and $10\,\Omega$ are connected respectively in cyclic order to form Wheatstone's network. The resistance that is to be connected in parallel with the resistance of $10\,\Omega$ to balance the network is _____ Ω. *[JEE Main 2020, 8 Jan, Mor.]*

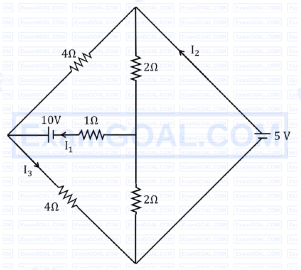
Q.15 Current measured by the ammeter (A) in the reported circuit when no current flows through $10\,\Omega$ resistance, will be _____ A. *[JEE Main 2022, 28 Jun, Mor.]*



Q.16 Consider a 72 cm long wire AB as shown in the figure. The galvanometer jockey is placed at P on AB at a distance x cm from A. The galvanometer shows zero deflection. The value of x, to the nearest integer, is _____. *[JEE Main 2021, 18 Mar, Eve.]*



Q.17 Refer to the figure given below. The values of I_1, I_2 and I_3 are: *[JEE Main 2026, 5 Apr, Mor.]*



(A) $I_1 = 2.5$ A, $I_2 = 1.875$ A, $I_3 = 1.875$ A

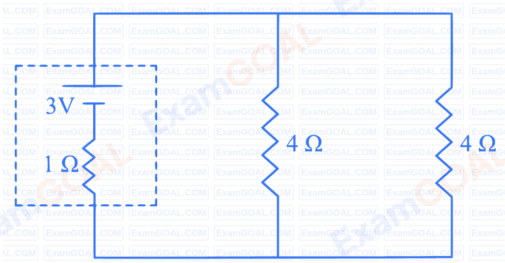
(B) $I_1 = 1.875$ A, $I_2 = 2.5$ A, $I_3 = 1.875$ A
(C) $I_1 = 1.875$ A, $I_2 = 1.875$ A, $I_3 = 2.5$ A
(D) $I_1 = 2.5$ A, $I_2 = 2.5$ A, $I_3 = 1.875$ A

Q.18 Two cells of emfs 1 V and 2 V and internal resistances $2\,\Omega$ and $1\,\Omega$, respectively, are connected in series with an external resistance of $6\,\Omega$. The total current in the circuit is I_1. Now the same two cells in parallel configuration are connected to same external resistance. In this case, the total current drawn is I_2. The value of $\left(\frac{I_1}{I_2}\right)$ is $\frac{x}{3}$. The value of x is _____. *[JEE Main 2025, 3 Apr, Eve.]*

Q.19 Two identical cells, when connected either in parallel or in series gives same current in an external resistance $5\,\Omega$. The internal resistance of each cell will be _____ Ω. *[JEE Main 2023, 31 Jan, Mor.]*

Q.20 In an electric circuit, a cell of certain emf provides a potential difference of 1.25 V across a load resistance of $5\,\Omega$. However, it provides a potential difference of 1 V across a load resistance of $2\,\Omega$. The emf of the cell is given by $\frac{x}{10}$ V. Then the value of x is _____. *[JEE Main 2021, 22 Jul, Eve.]*

Q.21 In the given circuit, the terminal potential difference of the cell is: *[JEE Main 2024, 8 Apr, Mor.]*



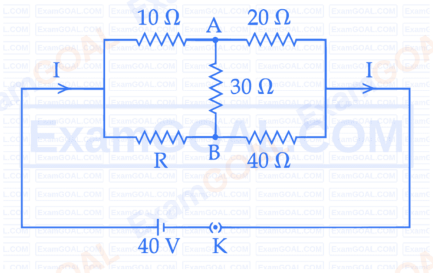
(A) 3 V (B) 4 V (C) 2 V (D) 1.5 V

Q.22 Two sources of equal emfs are connected in series. This combination is connected to an external resistance R. The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero, then the value of R will be: *[JEE Main 2022, 27 Jul, Mor.]*

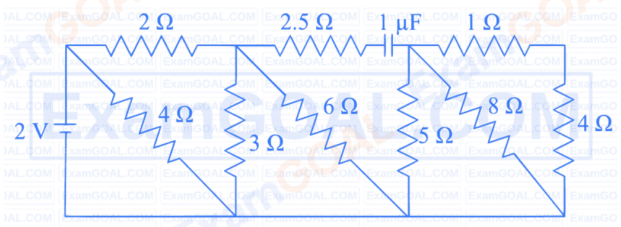
(A) $r_1 - r_2$ (B) $\frac{r_1 r_2}{r_1 + r_2}$
(C) $\frac{r_1 + r_2}{2}$ (D) $r_2 - r_1$

Q.23 A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by $5\,\Omega$. When a resistance of $15\,\Omega$ is used for shunting, null point moves to 300 cm. The internal resistance of the cell is _____ Ω. *[JEE Main 2023, 29 Jan, Eve.]*

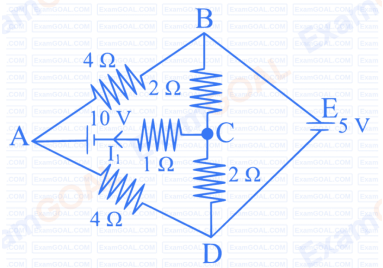
Q.24 The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A. *[JEE Main 2025, 28 Jan, Eve.]*



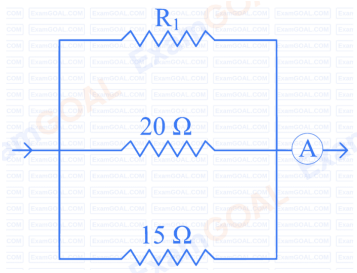
Q.25 The net current flowing in the given circuit is _____ A. *[JEE Main 2025, 22 Jan, Eve.]*



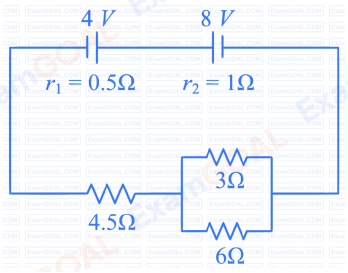
Q.26 The current flowing through the $1\,\Omega$ resistor is $\frac{n}{10}$ A. The value of n is _____. *[JEE Main 2024, 9 Apr, Mor.]*



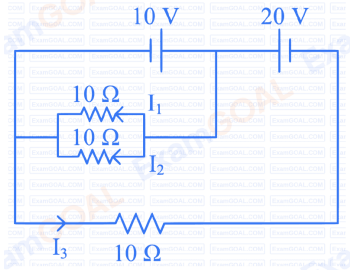
Q.27 In the given circuit, the current flowing through the resistance $20\,\Omega$ is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is _____ Ω. *[JEE Main 2024, 29 Jan, Eve.]*



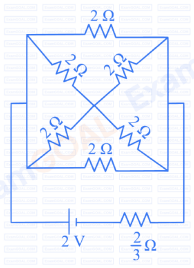
Q.28 In the circuit diagram shown in figure given below, the current flowing through resistance $3\,\Omega$ is $\frac{x}{3}$ A. The value of x is _____. *[JEE Main 2023, 11 Apr, Mor.]*



Q.29 In the given circuit, the value of $\left|\frac{I_1 + I_3}{I_2}\right|$ is _____. *[JEE Main 2023, 1 Feb, Eve.]*



Q.30 In the following circuit, the battery has an emf of 2 V and an internal resistance of $\frac{2}{3}\,\Omega$. The power consumption in the entire circuit is _____ W. *[JEE Main 2024, 31 Jan, Eve.]*



SELF-ASSESSMENT – LEVEL 3 SHEET

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 30	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S
Q.21 ☐C ☐W ☐S	**Q.22** ☐C ☐W ☐S	**Q.23** ☐C ☐W ☐S	**Q.24** ☐C ☐W ☐S
Q.25 ☐C ☐W ☐S	**Q.26** ☐C ☐W ☐S	**Q.27** ☐C ☐W ☐S	**Q.28** ☐C ☐W ☐S
Q.29 ☐C ☐W ☐S	**Q.30** ☐C ☐W ☐S		

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake
☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

Topics / questions I need to revise:

My action plan before the next sheet:

Teacher's remarks:

Student's signature: _____ Teacher's signature: _____